

Econometrics I

Lecture 13b: Duration Models

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- ▶ The outcome is a **length of time until something happens**: months until a subscriber cancels, days until a loan defaults, quarters until a firm exits, weeks until a job seeker is hired, years until a patent is cited.
- ▶ Two features make this its own subject:
 1. **Censoring**. When the data end, many spells are still running. You know the duration is *at least* c_i . Throwing those out, or treating them as completed, is wrong in opposite directions.
 2. **The question is about timing**. “Does a price increase raise churn?” means “does it raise the *rate* at which subscribers leave, given they are still here?” That object is the hazard, and it is what business users of these models actually want.
- ▶ Readings: Wooldridge Ch. 22; Kiefer (1988, *JEL*).

Survival and hazard functions

Let $T \geq 0$ be the duration, with CDF $F(t)$ and density $f(t)$.

$$S(t) = \mathbb{P}(T > t) = 1 - F(t)$$

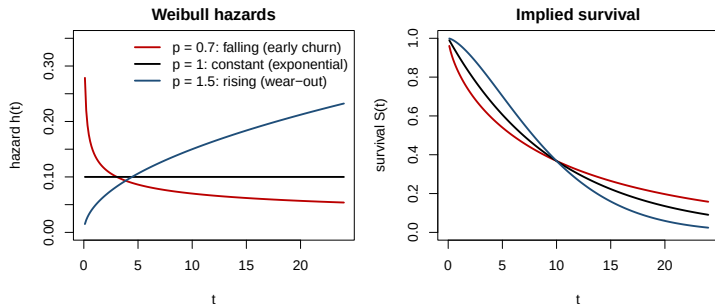
survival function: share still “alive” at t

$$h(t) = \lim_{\Delta \rightarrow 0} \frac{\mathbb{P}(t \leq T < t + \Delta \mid T \geq t)}{\Delta} = \frac{f(t)}{S(t)}$$

hazard: exit rate at t , given survival to t

- ▶ Since $f = -S'$, $h(t) = -\frac{d}{dt} \ln S(t)$, so $S(t) = \exp \left(- \int_0^t h(u) du \right) \equiv e^{-\Lambda(t)}$. The hazard determines everything: model it, and S , f , and $\mathbb{E}[T]$ follow.
- ▶ The hazard is what the decision-maker faces *now*: “of the subscribers still with us at month 6, what fraction leave in month 7?”
- ▶ Discrete time: $h_t = \mathbb{P}(T = t \mid T \geq t)$, $S(t) = \prod_{s \leq t} (1 - h_s)$: a sequence of binary choices, which is why a logit with time dummies is a duration model.

Duration dependence: what shape is the hazard?



- **Constant** hazard (exponential): memoryless; the expected remaining life never changes.
- **Falling** hazard: churn concentrated early (trial users), default risk highest in the first year of a loan.
- **Rising** hazard: wear-out; contract renewal dates; a job search whose reservation wage drifts down.

Running example: subscription churn

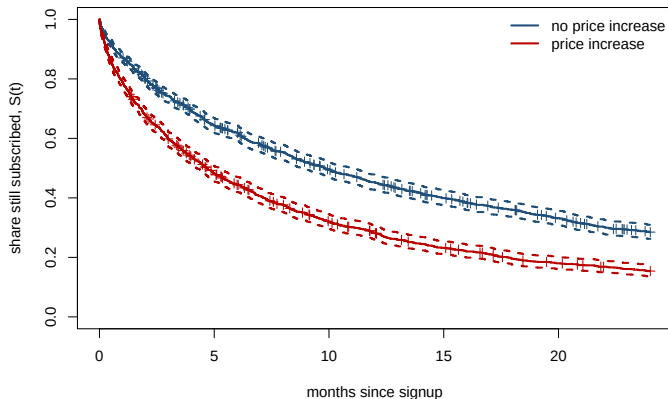
- ▶ 3,000 subscribers. Covariates: whether they experienced a **price increase** (half did) and their standardized **usage**.
- ▶ True hazard, Weibull with proportional covariate effects:

$$h(t \mid x) = \underbrace{p t^{p-1}}_{\text{baseline}} \cdot \exp(0.5 \cdot \text{price} - 0.4 \cdot \text{usage}), \quad p = 0.7.$$

A price increase multiplies the exit rate by $e^{0.5} \approx 1.65$ at every t . Falling baseline: early churn.

- ▶ We observe subscribers for 24 months; some also drop out of the panel early for unrelated reasons. About a quarter of spells are **right-censored**: we know they lasted at least c_i .
- ▶ Data: (t_i, d_i, x_i) with $t_i = \min(T_i, c_i)$ and $d_i = \mathbb{1}\{T_i \leq c_i\}$ an *event indicator*. That pair is the whole trick.

First look: Kaplan–Meier



Nonparametric survival by group, censored spells marked. At each exit time t_j , $\hat{S}(t) = \prod_{t_j \leq t} (1 - d_j/n_j)$: exits over the number *at risk*. Censored subscribers stay in the denominator until they leave the data and never enter the numerator.

The censored likelihood

Each subscriber contributes one of two things, exactly as in the Tobit:

$$\ell(\theta) = \sum_i \left[d_i \ln f(t_i | x_i; \theta) + (1 - d_i) \ln S(t_i | x_i; \theta) \right] = \sum_i \left[d_i \ln h(t_i | x_i; \theta) - \Lambda(t_i | x_i; \theta) \right]$$

- ▶ Completed spells ($d_i = 1$): the **density** at the observed duration.
- ▶ Censored spells ($d_i = 0$): the **probability** of lasting at least this long, $S(c_i)$.
- ▶ Second form uses $f = hS$ and $S = e^{-\Lambda}$: every spell pays the cumulative hazard for the time it was observed; completed spells also pay $\ln h$ at the moment of exit.
- ▶ Requires censoring to be **independent** of T given x : the data ending on 12/31 is fine; customers leaving the panel *because* they are about to churn is not (that is selection again).
- ▶ Everything from Week 12 applies: score, Hessian, sandwich standard errors, likelihood-ratio tests.

Parametric hazards: exponential and Weibull

Proportional hazards form: $h(t | x) = h_0(t) \exp(x' \beta)$, with a parametric baseline.

- ▶ **Exponential:** $h_0(t) = \lambda$. Then $\Lambda(t | x) = \lambda t e^{x' \beta}$, $\mathbb{E}[T | x] = e^{-x' \beta} / \lambda$. One parameter, no duration dependence.
- ▶ **Weibull:** $h_0(t) = \lambda p t^{p-1}$. $p < 1$ falling, $p = 1$ exponential, $p > 1$ rising.
- ▶ Interpretation of β_j : $\partial \ln h / \partial x_j$. e^{β_j} is the **hazard ratio**: the multiplicative effect on the exit rate, constant over t by assumption.
- ▶ Software often reports the **accelerated failure time** (AFT) form instead, $\ln T = \alpha + x' \gamma + \text{error}$: covariates stretch or compress time. For the Weibull the two are the same model with $\beta = -\gamma p$, and `survreg` in R uses AFT. Check which one you are reading before you interpret a sign.
- ▶ Wrong baseline shape biases $\hat{\beta}$ too, because the shape and the covariates compete to explain when exits happen. Next table.

Cox: leave the baseline alone

- ▶ Cox (1972): keep $h(t \mid x) = h_0(t) e^{x'\beta}$ but do not specify h_0 at all.
- ▶ At each exit time t_j , ask: of everyone still at risk, why was it subscriber j who left?

$$\mathbb{P}(j \text{ exits at } t_j \mid \text{one exit among the risk set } R_j) = \frac{h_0(t_j) e^{x'_j \beta}}{\sum_{k \in R_j} h_0(t_j) e^{x'_k \beta}} = \frac{e^{x'_j \beta}}{\sum_{k \in R_j} e^{x'_k \beta}}.$$

The baseline cancels. Multiply these over exit times: the **partial likelihood**. Maximize it.

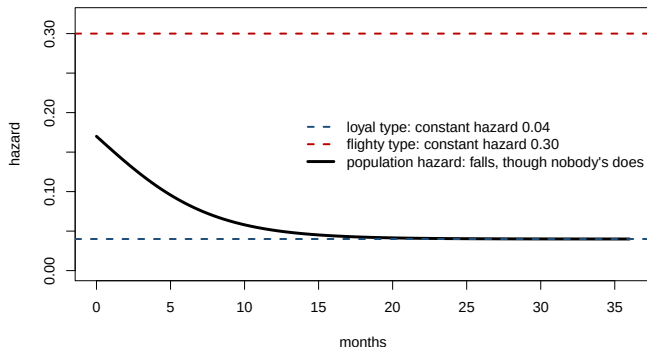
- ▶ Each term is a conditional logit (Week 12) over who in the risk set exits. Censored spells sit in risk sets and then drop out; they never contribute a numerator.
- ▶ You get $\hat{\beta}$ and hazard ratios without ever estimating h_0 ; $\hat{h}_0$ can be recovered afterwards (Breslow) if you need predicted survival.
- ▶ Cost: proportionality is now the only assumption, and it is testable (interact x with $\ln t$, or plot $\ln(-\ln \hat{S})$ by group; parallel lines mean proportional hazards).

Estimates on the simulated subscribers

	Cox PH	Weibull PH	Exponential PH	OLS $\ln T$, uncensored only	OLS $\ln T$, censored
Price increase (truth $\beta = 0.5$; HR = 1.65)	0.483 (0.042)	0.485 (0.042)	0.570 (0.042)	-0.409 (0.072)	-0.56 (0.06)
Usage (truth $\beta = -0.4$)	-0.384 (0.023)	-0.385 (0.023)	-0.457 (0.023)	0.213 (0.038)	0.38 (0.03)
Shape p (truth = 0.7)	—	0.70	1 (imposed)	—	—
Baseline hazard	unrestricted	Weibull	constant	(AFT, wrong sign)	(AFT, wrong sign)

- Cox and Weibull agree with the truth and with each other; the Weibull also recovers the shape.
- The exponential imposes $p = 1$ and gets the covariate effects wrong, not just the baseline.
- The two OLS columns are the regressions people run before they learn this: dropping censored spells selects on short durations; treating them as exits mislabels the longest spells. Both have the wrong sign convention as well (AFT: higher hazard means *shorter* $\ln T$).

Unobserved heterogeneity: the hazard that falls though nobody's does



- ▶ Two types with *constant* hazards. The flighty type leaves early; survivors are increasingly loyal. The population hazard falls: **spurious negative duration dependence**.
- ▶ So a falling estimated hazard has two readings: individuals become less likely to leave over time, or the composition of who is left changes. Data on spells alone cannot tell them apart.

What to do about heterogeneity

- ▶ **Condition on more.** Covariates absorb some of the heterogeneity; the rest is a frailty term ν_i multiplying the hazard, $h(t \mid x, \nu) = \nu h_0(t) e^{x'\beta}$.
- ▶ **Mixed proportional hazards.** Put a distribution on ν (gamma, usually) and integrate it out. Identified under proportionality with a covariate that shifts the hazard (Elbers–Ridder 1982), but fragile: h_0 and the distribution of ν are separated by functional form.
- ▶ **Repeated spells** per unit (several subscriptions, several loans) identify ν_i from within-unit variation, the panel logic of Week 8. Much stronger.
- ▶ **Even without a fix**, omitted ν biases $\hat{\beta}$ toward zero as t grows: high- ν units with hazard-raising x leave first, so the estimated hazard ratio drifts even when the true one is constant.
- ▶ Treat estimated duration dependence as descriptive unless a design separates it from sorting.

Design issues that come up every time

- ▶ **Left truncation (delayed entry).** You start observing subscribers who signed up before your data began. They are in the data only because they survived to the start. Condition on $T > \text{entry time}$: risk sets must exclude units before they enter. Software handles it if you tell it.
- ▶ **Time-varying covariates.** Price changes mid-spell. Split each spell into intervals with constant x (“counting-process” data); the likelihood in the second form on the earlier slide handles it directly. Cox handles it too.
- ▶ **Competing risks.** A subscriber can cancel, or upgrade, or die. Treating the other exits as censoring gives the cause-specific hazard, which is fine for “what raises cancellation?” but not for “what share will have cancelled by month 12?” (that needs the cumulative incidence).
- ▶ **Grouped time.** Monthly billing data records exits in months, not instants. Use the discrete-time model: a logit or complementary log-log on person-months with duration dummies. It is a panel logit, and it lets you use everything from Weeks 8 and 12.

The survival package ships with R:

```
library(survival)
km <- survfit(Surv(time, event) ~ price, data = d); plot(km)           # Kaplan-Meier
cox <- coxph(Surv(time, event) ~ price + usage, data = d)             # hazard ratios: exp(coef(cox))
wei <- survreg(Surv(time, event) ~ price + usage, data = d, dist = "weibull")
-coef(wei)[-1] / wei$scale; 1 / wei$scale                             # PH coefficients; shape p
```

Discrete time, with only fixest:

```
pm <- expand_to_person_months(d)                                       # one row per subscriber-month at risk
feglm(exit ~ price + usage | month, data = pm, family = binomial("cloglog"))
```

- ▶ The cloglog link makes the discrete-time model the exact grouped version of proportional hazards: coefficients are the same β . Month fixed effects are the baseline hazard.
- ▶ Python: lifelines (KM, Cox, Weibull) or the same person-month logit in pyfixest.

- ▶ Model the **hazard**: the exit rate given survival. It determines S , f , and $\mathbb{E}[T]$.
- ▶ Censored spells enter the likelihood through $S(c_i)$, completed ones through $f(t_i)$. Requires censoring independent of T given x .
- ▶ Kaplan–Meier to look; Weibull or exponential when you want the shape; Cox when you do not want to assume it. Read AFT output carefully.
- ▶ A falling hazard is either duration dependence or sorting. Spell data alone cannot say which.
- ▶ Monthly data plus `fixest`: a person-month cloglog with month dummies is a proportional-hazards model.